Lösung: Trigonometrische Gleichungen

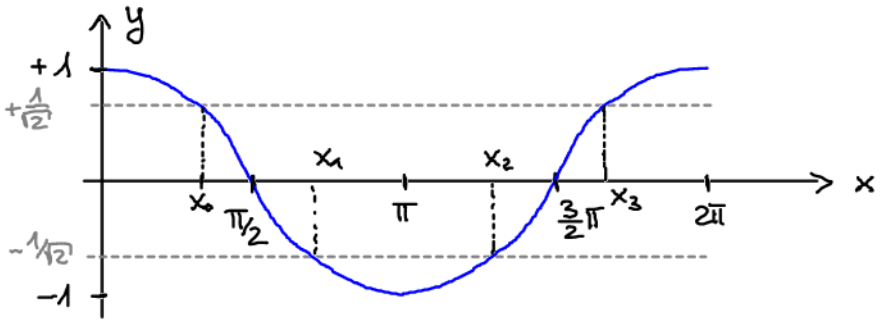
a) $\sin^2(x) + 2 \cdot \cos^2(x) = 3/2$

$\underbrace{\sin^2(x) + \cos^2(x)}_{=1} + \cos^2(x) = 3/2$

$1 + \cos^2(x) = 3/2 \quad \Rightarrow \quad \cos^2(x) = 3/2 - 1$

$\cos^2(x) = 1/2 \quad \Rightarrow \quad \cos(x) = \pm 1/\sqrt{2}$

$x_0 = \arccos\left(\pm 1/\sqrt{2}\right)$



$x_0 = \pi/4 \qquad (45°)$

$x_1 = \pi - \pi/4 = \frac{3}{4}\pi \qquad (135°)$

$x_2 = \pi + \pi/4 = \frac{5}{4}\pi \qquad (225°)$

$x_3 = 2\pi - \pi/4 = \frac{7}{4}\pi \qquad (315°)$

b) $2\sin(2x) - \sqrt{3} = 0$

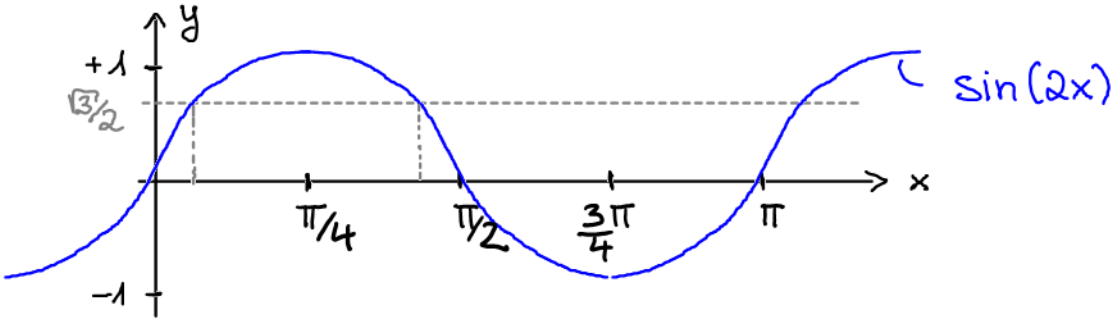
$$\sin(2x) = \frac{\sqrt{3}}{2}$$

$$2x_k = \arcsin\left(\frac{\sqrt{3}}{2}\right) \pm k \cdot 2\pi$$

$$x_k = \frac{1}{2}\arcsin\left(\frac{\sqrt{3}}{2}\right) \pm k \cdot \pi$$

$$x_{1k} = \frac{1}{2}\frac{\pi}{3} \pm k \cdot \pi \quad \wedge \quad x_{2k} = \frac{1}{2}\left(\pi - \frac{\pi}{3}\right) \pm k \cdot \pi$$

$$x_{1k} = \frac{\pi}{6} \pm k\pi \quad \wedge \quad x_{2k} = \frac{2}{6}\pi \pm k\pi$$



$x_{10} = \pi/6$ $(= 30°)$

$x_{20} = \pi/3$ $(= 60°)$

$x_{11} = 7/6\,\pi$ $(= 210°)$

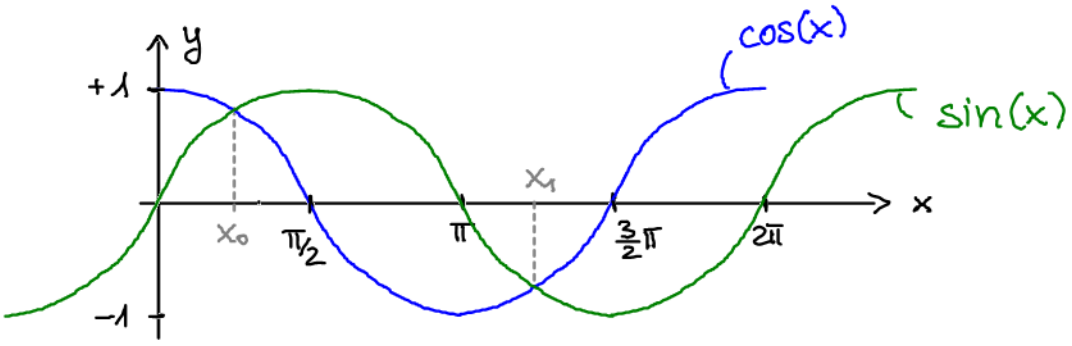
$x_{21} = 8/6\,\pi$ $(= 240°)$

c) $\sin(x) = \cos(x) \quad \Rightarrow \quad \tan(x) = 1$

$x_n = \underbrace{\arctan(1)}_{= \pi/4} + n \cdot \pi$

↑ da $\tan(x)$ eine Periode von π hat

$\underline{\underline{x_n = \pi\,(n + 1/4)}} \qquad (180° \cdot n + 45°)$



d) $\sin(x) \cdot \cos(x) = 0$

↑ ist erfüllt wenn $\sin(x) = 0$ oder $\cos(x) = 0$

Nullstellen von $\sin(x)$: $\quad x_n = k \cdot \pi$

Nullstellen von $\cos(x)$: $\quad x_n = k \cdot \pi + \pi/2$

→ Nullstellen von $\sin(x)\cos(x)$: $\quad \underline{x_n = k \cdot \pi/2}$

e) $\cos(2x) - \cos(x) = 0$

$(2\cos^2(x) - 1) - \cos(x) = 0$

$2\cos^2(x) - \cos(x) - 1 = 0$

Substitution: $z = \cos(x)$

$2z^2 - z - 1 = 0$

$z^2 - \frac{1}{2}z - \frac{1}{2} = 0$

$z_{1/2} = \frac{1}{2} \pm \sqrt{\frac{1}{4} + \frac{1}{2}} = \frac{1}{2} \pm \sqrt{\frac{3}{4}} = \frac{1 \pm \sqrt{3}}{2}$

Rücksubstitution: $x_n = \arccos(z) + 2\pi \cdot n$

$$x_n = \arccos\left(\frac{1 \pm \sqrt{3}}{2}\right) + 2\pi \cdot n$$